

Technical Compilation and Calculation Annex

Consolidated CBAM evidence dossier and preparation statements

Report report_aa · Release 5

AUTOMATED PREPARATION CHECKS
PASSED

CONFIDENTIAL - VERIFIER PREPARATION WORKSPACE

DOCUMENT BOUNDARY

This package supports preparation for independent verification. It is not a verification opinion, accreditation decision, customs decision, CBAM Registry submission or acceptance guarantee.

Control field			
Case ID	case_verifier_grade_fixture	Generated	2027-01-15T12:00:00.000Z
Ruleset	3.0.0	Source registry	CBAM-EU-2026.07.16
Root hash	7e12ba9592177cc10e0e4541bb103905e0345303fa794949881b86ec2ab48f61	Verifier status	NOT_REVIEWED

Product Scope - Controlled identity

Field	
Importer	CBAMValid Importer B.V.
EORI	NL123456789AB
Exporter / operator	Verified Steel Operator GmbH
Installation	Verified Integrated Steel Installation
Country	DE
Production route	Blast Furnace Route (BF-BOF)
Reporting period	2026-ANNUAL

Product Scope - Goods population

Good							
1	72011011	IRON_AND_STEEL	60	0.6	72	1.2	0.06
2	72011019	IRON_AND_STEEL	40	0.4	48	1.2	0.06

Product Scope - Scope conclusion

AUTOMATED READINESS
READY_FOR_INDEPENDENT_VERIFICATION

Product Scope - Legal basis

Source			
REG_2023_956	32023R0956	2023-05-17	CBAM framework; Annex I goods scope; Annex IV embedded-emissions calculation framework

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Source			
REG_2025_2083	32025R2083	2025-10-20	50 tonne annual mass threshold for cement, fertilisers, iron and steel, and aluminium; definitive-period declaration and certificate obligations; actual values require independent verification
IMPL_2025_2546	32025R2546	2026-01-01	reasonable assurance and risk-based verification; 5 percent per-good materiality levels; electronic verification report template
IMPL_2025_2547	32025R2547	2026-01-01	functional units and production processes; monitoring plan minimum elements; sector-specific system boundaries and embedded-emissions methods
IMPL_2025_2548	32025R2548	2026-01-01	calculation and publication of CBAM certificate prices
DEL_2025_2551	32025R2551	2026-01-01	accreditation and competence of verifiers; verification planning, evidence and site-visit framework; oversight and independence

CN Code - Classification register

Good				
1	72011011	IRON_AND_STEEL	11111111-1111-4111-8111-222222222222	11111111-1111-4111-8111-444444444444
2	72011019	IRON_AND_STEEL	11111111-1111-4111-8111-222222222222	11111111-1111-4111-8111-444444444444

CN Code - Classification boundary

This document records the classification supplied in the sealed case. It does not replace a binding customs classification decision. Any disputed or uncertain CN code remains subject to customs review.

Checklist - Monitoring-plan requirements

Require ment			
MP-01	Monitoring-plan date, version and responsible operator identity	DOCUMENTE D	Case version 1; owner recorded
MP-02	Installation description and production processes	DOCUMENTE D	Blast Furnace Route (BF-BOF)
MP-03	CN-coded goods and functional production units	DOCUMENTE D	2 good(s) recorded
MP-04	System boundaries and directly connected processes	DOCUMENTE D	Coke preparation, sinter plant, blast furnace, basic oxygen furnace, casting and finishing operations within the controlled installation boundary.
MP-05	Source streams, emission sources and measurement methods	DOCUMENTE D	0 precursor(s); direct and electricity inputs recorded
MP-06	Data-source hierarchy, corroboration and control activities	DOCUMENTE D	4 approved and clean evidence file(s)
MP-07	Calculation methods, conversions and rounding	DOCUMENTE D	8 hashed trace node(s)
MP-08	Goods allocation method and reconciliation	DOCUMENTE	Allocation delta 0

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Require ment			
		D	
MP-09	Precursor scope and embedded emissions treatment	DOCUMENTE D	Accepted no-precursor decision
MP-10	Methodology deviations, estimates and technical justifications	DOCUMENTE D	2/2 methodology decision(s) accepted
MP-11	Per-good materiality assessment	DOCUMENTE D	5% per-good specific-emissions threshold
MP-12	Evidence integrity and immutable calculation provenance	DOCUMENTE D	Calculation root 7e12ba9592177cc10e0e4541bb103905e0345303fa794949881b86ec2ab48f61

Checklist - Automated control findings

Rule				
QC_00	Operator, installation and boundary identity	PASS	No automated finding	REM_COMPLETE_C ASE_IDENTITY
QC_01	EORI format and evidence	PASS	No automated finding	—
QC_02	Definitive-period reporting year	PASS	No automated finding	REM_CORRECT_RE PORTING_YEAR
QC_03_0	Good 1 CN code and evidence	PASS	No automated finding	—
QC_04_0	Good 1 production and evidence	PASS	No automated finding	—
QC_05_0	Good 1 sector	PASS	No automated finding	REM_SELECT_SUPP ORTED_SECTOR
QC_03_1	Good 2 CN code and evidence	PASS	No automated finding	—
QC_04_1	Good 2 production and evidence	PASS	No automated finding	—
QC_05_1	Good 2 sector	PASS	No automated finding	REM_SELECT_SUPP ORTED_SECTOR
QC_05A	Allocation, evidence and reconciliation	PASS	No automated finding	—
QC_06	directEmissions value, unit and evidence	PASS	No automated finding	—
QC_07	electricityConsumed value, unit and evidence	PASS	No automated finding	—
QC_08	gridEmissionFactor value, unit and evidence	PASS	No automated finding	—
QC_09	Precursor scope	PASS	No automated finding	REM_CONFIRM_PRE CURSOR_SCOPE
QC_10	Evidence integrity	PASS	No automated finding	—
QC_11	Carbon price records	NOT_APPLICA BLE	No automated finding	—

Checklist - Evidence coverage

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Measure	
Registered evidence files	4
Approved and malware-clean files	4
Linked input fields	18
Linked calculation nodes	5
Coverage rate	100%
Duplicate hashes	0

Monitoring Plan - Installation and boundary

Coke preparation, sinter plant, blast furnace, basic oxygen furnace, casting and finishing operations within the controlled installation boundary.

Production route: Blast Furnace Route (BF-BOF)

Monitoring Plan - Minimum monitoring-plan elements

Require ment			
MP-01	Monitoring-plan date, version and responsible operator identity	DOCUMENTE D	Case version 1; owner recorded
MP-02	Installation description and production processes	DOCUMENTE D	Blast Furnace Route (BF-BOF)
MP-03	CN-coded goods and functional production units	DOCUMENTE D	2 good(s) recorded
MP-04	System boundaries and directly connected processes	DOCUMENTE D	Coke preparation, sinter plant, blast furnace, basic oxygen furnace, casting and finishing operations within the controlled installation boundary.
MP-05	Source streams, emission sources and measurement methods	DOCUMENTE D	0 precursor(s); direct and electricity inputs recorded
MP-06	Data-source hierarchy, corroboration and control activities	DOCUMENTE D	4 approved and clean evidence file(s)
MP-07	Calculation methods, conversions and rounding	DOCUMENTE D	8 hashed trace node(s)
MP-08	Goods allocation method and reconciliation	DOCUMENTE D	Allocation delta 0
MP-09	Precursor scope and embedded emissions treatment	DOCUMENTE D	Accepted no-precursor decision
MP-10	Methodology deviations, estimates and technical justifications	DOCUMENTE D	2/2 methodology decision(s) accepted
MP-11	Per-good materiality assessment	DOCUMENTE D	5% per-good specific-emissions threshold
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Monitoring Plan - Input hierarchy

Input					
Direct emissions	80	tCO2e	PRIMARY	11111111-1111-4111-8111-5555555555	Documented direct measurement and reconciled production ledger
Electricity	100	MWh	PRIMARY	11111111-1111-4111-8111-3333333333	Documented direct measurement and reconciled production ledger
Grid emission factor	0.4	tCO2e/MWh	PRIMARY	11111111-1111-4111-8111-3333333333	Documented direct measurement and reconciled production ledger

Process Map - End-to-end process

Stage		
1	Identify installation, operator, CN-coded goods and reporting period	CASE and GOODS registers
2	Define production processes, source streams, boundaries and functional units	Monitoring Plan and System Boundary
3	Capture direct, electricity and precursor data with approved evidence	INPUTS, PRECURSORS and EVIDENCE
4	Apply unit conversions, allocation and embedded-emissions formulae	CALCULATION_TRACE and root hash
5	Run evidence, methodology, reconciliation and integrity controls	QUALITY_CONTROLS and MONITORING_PLAN
6	Apply per-good 5% materiality reference for verifier planning	GOODS materiality columns
7	Freeze, hash, sign and commit immutable package	Manifest, KMS signature and package hash
8	Independent accredited verifier performs review and records opinion	VERIFIER_SIGN_OFF — initially NOT_REVIEWED

System Boundary - Declared system boundary

BOUNDARY

Coke preparation, sinter plant, blast furnace, basic oxygen furnace, casting and finishing operations within the controlled installation boundary.

System Boundary - Sector methodology

Sector			
Iron and Steel	IN_FORCE	Coke oven, sinter or pellet preparation, blast furnace, basic oxygen furnace, direct reduction, electric arc furnace, casting and finishing processes within the installation boundary.	Mass balance and production ledger; Fuel and reducing-agent consumption; Precursor embedded emissions; Waste-gas and transferred-heat treatment; Allocation to CN-coded goods

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Methodology - Decision register

Topic					
PRECURSOR_SC OPE	No qualifying precursor line is declared for the fixture goods and controlled route.	The controlled bill of materials and production route contain no separate precursor goods requiring an additional precursor line in this fixture.	Regulation (EU) 2023/956 Annex IV and Commission Implementing Regulation (EU) 2025/2547; supported by the controlled bill of materials.	ACCEPTED	11111111-1111-4111-8111-555555555555
GOODS_EMISSIONS_ALLOCATION	Allocate installation emissions using documented product mass shares of 0.6 and 0.4.	The two goods are produced in the same controlled period and the reconciled product ledger provides complete mass shares summing to one.	Commission Implementing Regulation (EU) 2025/2547 allocation and monitoring-plan requirements; controlled production ledger.	ACCEPTED	11111111-1111-4111-8111-444444444444

Calculation Annex - Engine identity

Control	
Ruleset	EU-CBAM-DEFINITIVE-2026
Engine version	3.0.0
Calculation root hash	7e12ba9592177cc10e0e4541bb103905e0345303fa794949881b86ec2ab48f61
Allocation share total	1
Allocation reconciliation delta	0

Calculation Annex - Formula trace

Formula				
CBAM_INDIRECT_EMISSIONS	40	tCO ₂ e	6a2647d365280b1a97c749006dfaa066e9374f7534b8c11cafe32b791c010350	None
CBAM_PRECURSOR_EMISSIONS_SUM	0	tCO ₂ e	5553fb3ff7a2d2d3f04af35bdc6b6f4c64f77ee3359321812bc30163f210d3c86	None
CBAM_TOTAL_EMBEDDED_EMISSIONS	120	tCO ₂ e	cc192e8f6a78218e061c33a7c0a99f82059fba5ee1def6dd1735ccabc099b2d2	None
CBAM_GOOD_EMISSIONS_ALLLOCATION_1	1.2	tCO ₂ e/t	7aba824be81004f963dad9b979458428808b42ab7d4f48dbfcae56a4f661fd5ba	None
CBAM_GOOD_EMISSIONS_ALLLOCATION_2	1.2	tCO ₂ e/t	2d6d72ee83e4cd16157fe19d8f21971c0c6b8e36076c5bbbf42403b5a39f5a9b	None
CBAM_GOODS_ALLOCATION_RECONCILIATION	0	tCO ₂ e	274077c9bd67745642e0aaf8506b146e80db2fca464cbcbf55485d95475c2fd0	None
CBAM_AGGREGATE_SPECIFIC_EMBEDDED_EMISSIONS	1.2	tCO ₂ e/t	61fff4226612cedc54b2dfd528abe1b84fd63436a2cef5e27dc6b2f41761f2d0	The aggregate intensity is a portfolio diagnostic; reportable values are the per-good intensities.
CBAM_CARBON_PRICE_REDUCTION_RECORD_TOTAL	0	certificate-equivalent	13eee823710e59782162be2fd529ca7dd9bb3ac7b195d5d21486017266c848ea	None

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Calculation Annex - Per-good reconciliation

Good							
1	72011011	IRON_AND_STEEL	60	0.6	72	1.2	0.06
2	72011019	IRON_AND_STEEL	40	0.4	48	1.2	0.06